

Module 1: Dataset discovery, acquisition, and quality control

Instructional time: 45 min. Bloom: Remember, Understand, Apply. Notebook: notebooks/01_discovery_and_qc.ipynb

1. Overview

Module 1 is the entry point to the CFDE lung transcriptomics course. It covers how the required datasets are identified and retrieved, how a local copy is verified against the expected inventory, and how single-nucleus quality control and preprocessing are applied so that four HuBMAP lung donors are ready for integration in Module 2.

Learning objectives. After this module you will be able to:

- Use the CFDE discovery surfaces to locate lung datasets across Common Fund resources, and explain how the discovery layer relates to the DCC portals that serve the files.
- Resolve a HuBMAP dataset record to a downloadable release, and cite the correct record for it.
- Confirm that a local copy of the cohort is complete before analysis begins.
- Apply and justify single-nucleus quality-control thresholds, then run a standard normalize, highly variable gene, PCA and UMAP preprocess.

Downloading the full course data set takes about 814.87 MB. Module 1 itself uses the four HuBMAP single-nucleus matrices; the remainder is needed by Modules 3 and 4.

2. Before you begin

Prerequisites. Basic Python / AnnData. CFDE Workbench vs DCC portals (screencast).

Cohort. Donor_1 (37), Donor_2 (25), Donor_3 (56.8), Donor_4 (52.96).

Run order. Run the analysis sections in order; the notebook carries state between cells.

Biological question

Is the HuBMAP lung snRNA-seq cohort complete and verifiable locally, what role does each file serve, and can nuclei from each donor be prepared so that the cross-donor analysis in Module 2 rests on documented quality-control decisions?

Module 1 answers that question. It does not integrate donors, call differential expression, or leave HuBMAP for other ecosystems. Those steps begin in Modules 2 and 4.

HuBMAP teaching cohort

Four donors selected for comparative value across age, sex and reported race and ethnicity. Donor_2 additionally carries a paired single-nucleus multiome dataset, which is the basis of Module 3.

Donor	Age	Sex	Primary block	Processed release	Also provides
Donor_1	37	Male	HBM347.VDHS.379	HBM675.RVGB.258	

Donor_2	25	Female	HBM473.NKMR.872	HBM362.DZVK.533	multiome HBM828.GPVG.252
Donor_3	56.8	Female	HBM484.RGWP.797	HBM874.PWHS.622	
Donor_4	52.96	Male	HBM379.QCJT.435	HBM735.NMFW.852	

Three identifiers do three different jobs, and confusing them is the most common way to lose time here. **Cite** the primary block and its DOI ([10.35079/](https://doi.org/10.35079/) followed by the primary ID). **Download** the processed release, which is the openly available derivative. **Name the local folder** after the primary block, because that is the identifier `config/paths.yaml` resolves against.

Each donor has other tissue collections in HuBMAP; the blocks above are the lung single-nucleus datasets this course uses.

Directory layout

```

CFDE_Training_Final/
README.md           quick start and requirements
environment.yml      conda environment (CFDE_lung_env)
requirements.txt     pip alternative
config/paths.yaml   all paths, thresholds and cohort identifiers
notebooks/          Modules 1 to 4
scripts/            analysis package
docs/               the module documents, Modules 0 to 5
outputs/            committed tables, figures and reports
data/
genesets/           shipped with the package (2.17 MB)
reference/           shipped: orthologs, ERCC expected fold changes
source/             YOU DOWNLOAD INTO HERE (Appendix A)
HUBMAP/
  Donor_1/snRNAseq/HBM347.VDHS.379/{raw_expr.h5ad, HBM347.VDHS.379.tsv}
  Donor_2/snRNAseq/HBM473.NKMR.872/{raw_expr.h5ad, HBM473.NKMR.872.tsv}
  Donor_2/SNAREseq/HBM828.GPVG.252/multiome_mofa.hdf5
  Donor_3/snRNAseq/HBM484.RGWP.797/{raw_expr.h5ad, HBM484.RGWP.797.tsv}
  Donor_4/snRNAseq/HBM379.QCJT.435/{raw_expr.h5ad, HBM379.QCJT.435.tsv}
GTEX/
GEO/GSE150910/
OSDR/OSD-248/OSD-248_metadata_GLDS-248-ISA/
processed/          written by Module 1, not downloaded

```

Option 1: minimum download, recommended

Approximately **814.87 MB** across 20 files. This is the recommended option and it is sufficient to reproduce every result in the course. Every file is openly accessible without an account, and each one is resolved from `config/paths.yaml`, so nothing here is optional for the default path. Appendix A gives the terminal commands.

Resource	File	What it provides	MB
HuBMAP Donor_1	raw_expr.h5ad	Raw counts matrix for QC	68.57
HuBMAP Donor_1	HBM347.VDHS.379.tsv	Block metadata for provenance and inventory	0.02
HuBMAP Donor_2	raw_expr.h5ad	Raw counts matrix for QC	104.55
HuBMAP Donor_2	HBM473.NKMR.872.tsv	Block metadata	0.02

HuBMAP Donor_3	raw_expr.h5ad	Raw counts matrix for QC	150.92
HuBMAP Donor_3	HBM484.RGWP.797.tsv	Block metadata	0.02
HuBMAP Donor_4	raw_expr.h5ad	Raw counts matrix for QC	70.56
HuBMAP Donor_4	HBM379.QCJT.435.tsv	Block metadata	0.02
HuBMAP multiome	multiome_mofa.hdf5	MOFA factors and variance, the Module 3 default path	140.78
GTEX v10	gene_tpm_v10_lung.gct.gz	Expression levels for the HuBMAP to GTEX bridge	77.60
GTEX v10	gene_reads_v10_lung.gct.gz	Counts for the aging differential expression	32.49
GTEX v10	GTEX_Analysis_v10_Annotations_SampleAttributesDS.txt	Sample metadata: tissue, ischemic time	38.50
GTEX v10	GTEX_Analysis_v10_Annotations_SubjectPhenotypesDS.txt	Subject metadata: age bracket, sex	0.02
GEO GSE150910	GSE150910_gene-level_count_file.csv.gz	Counts for the IPF versus control contrast	8.12
GEO GSE150910	GSE150910_series_matrix.txt.gz	Sample annotations and phenotype	0.01
OSDR OSD-248	GLDS-248_rna_seq_RSEM_Unnormalized_Counts_rRNArm_GLbulkRNAseq.csv	RSEM counts, rounded for pydeseq2	10.90
OSDR OSD-248	GLDS-248_rna_seq_differential_expression_rRNArm_GLbulkRNAseq.csv	GeneLab differential expression, used as an independent cross-check	111.54
OSDR OSD-248	GLDS-248_rna_seq_contrasts_GLbulkRNAseq.csv	Contrast definitions naming the flight and ground arms	0.01
OSDR OSD-248	OSD-248_metadata_GLDS-248-ISA/s_OSD-248.txt	ISA sample metadata: arm, duration, euthanasia	0.06
OSDR OSD-248	OSD-248_metadata_GLDS-248-ISA/a_OSD-248_transcription-profiling_rna-sequencing-(rna-seq)_Illumina NovaSeq 6000.txt	ISA assay metadata: library prep, sequencer	0.16

Two further OSD-248 files are small and worth taking with the set: [GLDS-248_rna_seq_SampleTable_GLbulkRNAseq.csv](#) (GeneLab sample table) and [OSD-248_metadata_GLDS-248-ISA/i_Investigation.txt](#) (ISA investigation metadata, 0.02 MB).

Three points that decide whether Module 4 runs. **GTEx supplies both a counts matrix and a TPM matrix, and both are required:** the aging differential expression is computed on `gene_reads`, while `gene_tpm` is used only for the expression-level comparison against HuBMAP pseudobulk. **The two GTEx annotation tables assign donors to age brackets**, so the aging contrast cannot be defined without them. And **the metadata files are not optional extras**: the OSD-248 ISA tables are what identify which samples are flight and which are ground.

Option 2: full download, optional

A complete local mirror of every resource this course touched is roughly **65 GB**, and almost all of that mass is unused by Modules 1 to 4. Take this option only if you want offline Azimuth labels, the optional Module 3 MuData views, the spare HuBMAP blocks, or a second spaceflight study to explore.

HuBMAP delivers a processed dataset as a directory rather than a single file, so each block folder also contains `expr.h5ad`, `secondary_analysis.h5ad`, `scvelo_annotated.h5ad` and portal QC output. For Donor_1 the teaching matrix is 68.57 MB and its `secondary_analysis.h5ad` alone is 721 MB. Each donor also has alternate lung blocks that this course does not use, and the Donor_2 multiome has a full MuData object of about 14.7 GiB that only Module 3's optional path reads. Learners do not download that MuData file.

The one file worth weighing is `secondary_analysis.h5ad`, which carries the Azimuth cell-type labels. The pipeline does not require it: labels come from the shipped cache, or are read directly from the remote file when `module2.labels.fetch_remote` is set to `true`. Downloading roughly 720 MB per donor for a single annotation column is exactly the kind of decision this course asks you to make deliberately rather than by default.

What the full mirror adds, over and above the minimum:

Addition	Approximate size	Why you might want it
<code>secondary_analysis.h5ad</code> for the four teaching blocks	5,669.9 MB	Offline Azimuth labels instead of the shipped cache or a remote read
<code>expr.h5ad</code> and <code>scvelo_annotated.h5ad</code> per block	varies	Portal processed expression and RNA velocity objects
15 alternate HuBMAP blocks across the four donors	25.4 GB of <code>raw_expr</code> plus <code>secondary_analysis</code> , more with the per-block extras	Spare lung blocks this course does not teach
<code>secondary_analysis.h5mu</code> for the Donor_2 multiome	14.7 GiB	The full MuData views behind Module 3's MOFA object
Donor_2 multiome portal cluster PDFs	1.2 MB	Portal cluster maps for RNA, ATAC and combined
GTEx v11 lung TPM and reads	124.5 MB	A newer GTEx release, not wired into <code>config/paths.yaml</code>
OSD-248 VST counts and sample table	32.9 MB	Portal companions to the RSEM counts
NASA OSDR OSD-464 , 8 files	55.7 MB	A second rodent spaceflight study, described below

OSD-464 is available but not wired. It carries the same eight file types as OSD-248, counts, DE, contrasts, sample table, VST counts and three ISA tables, and no notebook path reads it. It is here for exploration, and one finding is worth stating before you spend time on it: OSD-464 carries the same ERCC spike-in mix confounding with study arm that OSD-248 does, so substituting it does not resolve that limitation.

3. Getting the data

Discovery and retrieval are the first methodological decisions in this course, not preliminaries to it. The dataset identifier you find through a discovery surface is frequently not the record you can download, and it is frequently not the record you should cite. This section establishes those distinctions and gives the operational path for each of the four resources the course uses.

The CFDE discovery layer and the DCC data portals

The Common Fund Data Ecosystem provides a cross-resource discovery layer: a shared metadata model (C2M2), search surfaces built on that model, and an interoperability context that makes a single query meaningful across repositories that were built independently. The Data Coordinating Centers, HuBMAP and GTEx among them, hold and serve the data files themselves.

Keeping those roles distinct matters practically. Searching for analysis-ready matrices on the CFDE surfaces returns dataset records rather than files; searching a DCC portal without the discovery layer means knowing in advance which repository to search. The productive path uses both: CFDE to establish what exists across the ecosystem, the DCC portal to obtain it.

Locating lung datasets through CFDE surfaces

The CFDE Workbench and Data Matrix accept a tissue-level query and return the repositories holding matching data, with HuBMAP and GTEx both represented for lung. This is the appropriate first step and is sufficient for a small, hand-selected cohort such as the one used here.

The teaching cohort in this course was selected manually for comparative value: four donors spanning 25 to 57 years, two female and two male, across three race and ethnicity groups. That is a pedagogical selection, not a production sampling strategy, and it is stated as such.

Programmatic discovery through the SmartAPI registry

Manual selection does not scale. The programmatic path matters at the point where a cohort is defined by criteria rather than by hand, and the registry is where it begins:

```
https://smart-api.info/api/query?q=hubmap&fields=info.title,info.version,servers
```

That query returns nine registered HuBMAP entries, including the Search, Entity, Spatial and Ontology APIs used below.

A registry records what a provider registered, not what is currently served. The registered GTEx entry points at dev.gtexportal.org, a host with no DNS record, while the production API at gtexportal.org/api/v2/ responds normally. Registry entries are a starting point to be verified, not an authority to be trusted, and confirming an endpoint resolves before building against it costs seconds. Date-stamp any such observation, since providers do correct their entries.

The registry is how you find an API; the API is how you resolve a dataset identifier to a file URL. Appendix A carries that path end to end as terminal commands: the SmartAPI query, the Entity API call that turns a HuBMAP identifier into the UUID an asset URL needs, and a retrieval command for every file in the minimum download. Working through it once is the difference between knowing that a programmatic path exists and being able to build one.

Access tiers: controlled primary data and public processed derivatives

HuBMAP applies two access tiers to single-nucleus data, and the distinction is a data-governance concept rather than a technical obstacle.

Primary datasets contain human genetic sequence. They are published, discoverable and citable, and they are held under controlled access consistent with NIH policy for human genomic data. **Processed derivatives** are generated for each primary by HuBMAP's central pipeline, carry no human genetic sequence, and are openly available. The two are distinct records with distinct HuBMAP identifiers.

Discovery surfaces return the primary. Retrieval uses the derivative. A researcher who does not expect this concludes that HuBMAP single-nucleus data is unavailable for open reuse, which is incorrect.

The traversal from one to the other is a five-step query against the Search API: retrieve the primary by identifier, read its `data_access_level`, read its `descendants`, filter those to `data_access_level == "public"`, and take the most recently published. For this cohort:

Donor	Primary (controlled)	Processed derivative (open)	raw_expr.h5ad
Donor_1	HBM347.VDHS.379	HBM675.RVGB.258	68.6 MB
Donor_2	HBM473.NKMR.872	HBM362.DZVK.533	104.6 MB
Donor_3	HBM484.RGWP.797	HBM874.PWHS.622	150.9 MB
Donor_4	HBM379.QCJT.435	HBM735.NMFW.852	70.6 MB

Donor_1 has two published derivatives, from 2021 and 2023, both serving an equivalent matrix. This course pins the 2023 release explicitly. Pinning a release and recording which one was used is the difference between a reproducible cohort definition and one that silently changes when a repository republishes.

Citation follows the primary, not the derivative. DOIs are registered on the primary datasets; the processed derivatives carry none. A manuscript therefore cites the primary DOI and states which processed release was analyzed. This package records both, as `primary_doi` and `processed_id` in `config/paths.yaml`, so the provenance chain is recoverable from the configuration alone.

Retrieving the files

Assets are addressed by the derivative's UUID:

<https://assets.hubmapconsortium.org/<uuid>/<filename>>

Donor	Processed derivative	UUID
Donor_1	HBM675.RVGB.258	171bd6114652ce2a1ff0e4757b5eae9
Donor_2	HBM362.DZVK.533	4c031c4aba3b656a07a5ab14a305cbba
Donor_3	HBM874.PWHS.622	ab3942cee59a994d73f6f89f8f5eb84a
Donor_4	HBM735.NMFW.852	f5ce0bfcef8d17a18e0a3911fb2b2726

The pipeline begins from raw counts, so one matrix per donor is required. Appendix A gives the retrieval command for each.

The full course requires about 814.87 MB across four resources, itemized under "Option 1: minimum download, recommended" above. The four HuBMAP matrices are 394.6 MB of that, the Donor_2 multiome MOFA object is 140.8 MB, and GTEx lung TPM is 77.6 MB.

Select the smallest artifact that answers the question. The `secondary_analysis.h5ad` release for these same four donors totals 5,669.9 MB, a factor of 14.4 larger, and the only element this pipeline requires from it is a cell-type label column that can be read directly from the remote file. Download volume is a design decision with real consequences for who can take a course or reproduce an analysis.

One operational note. The HuBMAP assets server does not implement HEAD, so a header-only request reports an error for files that download normally. Verify availability with a ranged request rather than a header request.

Access patterns differ by repository

The four resources in this course expose data through four different mechanisms:

Resource	Retrieval mechanism
HuBMAP	REST API with an Elasticsearch query body, then assets addressed by UUID
GTEX	No API for the bulk matrices; object storage over HTTPS
GEO	Directory listing served over HTTPS
NASA OSDR	Study-scoped REST listing, then a download endpoint with query parameters

This heterogeneity is the practical reality of multi-resource work and the reason a discovery layer has value. It also argues for naming exact files rather than pointing at a study: the OSD-248 listing contains 1,313 files including trimmed FASTQs of several gigabytes each, of which Module 4 requires two.

Verifying the local copy

Confirm that every expected file is present and matches its expected size, and report anything absent by name. A silently missing donor propagates into an integration step and is far more expensive to diagnose there than at acquisition. The notebook performs this check against the download manifest, and [scripts/verify_access.py](#) performs the equivalent check against the live URLs.

One format note worth anticipating: these four donors were deposited across a period in which the AnnData categorical encoding changed, so Donor_1 stores categorical columns under an older layout than the other three. Code that assumes a single layout will fail on part of the cohort.

4. Methods

Parameters for each step are the live values in [config/paths.yaml](#) under [module1](#) (and related keys); the per-step tables are collected in "Appendix B: parameters" at the end of this companion, numbered to match the subsections below. They are also stamped into [outputs/tables/module1_methods_parameters.tsv](#) when the notebook runs. Script paths are relative to the project root.

Each Methods step below corresponds to a numbered section of [notebooks/01_discovery_and_qc.ipynb](#), so the document and the notebook can be read side by side:

Methods step	Notebook section
1. Identify and place teaching files	1. Identify and place teaching files
2. Inventory and measured-size verification	2. Inventory and measured-size verification
3. Provenance and access table	3. Provenance and access table
4. Composition gate	4. Composition gate
5. QC filters	5. QC filters and preprocessing
6. Preprocess, teaching subsample, neighbors, and UMAP	5. QC filters and preprocessing

Steps 5 and 6 share one notebook section because a single call applies the filters and the preprocessing together. Notebook section 0 is environment setup, covered in Module 0, and the unnumbered "Discovery" section carries the CFDE and DCC framing from "Getting the data" above.

1. Identify and place teaching files

Script. Manual portal download guided by `config/paths.yaml` (`module1.block_id` / `module2.inputs`); layout helpers in `scripts/common/paths.py` (`snrna_block_dir`, `teaching_inputs` via `scripts/nb01_discovery/verify.py`).

What. For each teaching donor, locate the public processed HuBMAP product that carries `raw_expr.h5ad` (Salmon counts-like matrix) and the block metadata `.tsv`. Place files under `data/source/HUBMAP/<Donor>/snRNAseq/<primary_id>/` using the primary block ID as the folder name (as in the learner manifest). Record `processed_id` and UUID in config so the bytes on disk are traceable even when the citation DOI points at the protected primary.

Parameters: see Appendix B, section 1.

Why. Primary datasets with human genetic sequence are protected. Anonymous learners download public processed descendants. Keeping the folder named by the primary ID preserves the citation target while `processed_id` / UUID records what was fetched. Extra HuBMAP blocks under a donor are ignored unless listed in `module2.inputs`.

2. Inventory and measured-size verification

Script. `scripts/nb01_discovery/inventory.py` (`inventory_all_donors`, `summarize_inventory`, `write_inventory_outputs`); `scripts/nb01_discovery/verify.py` (`verify_teaching_files`, `learner_download_manifest`, `write_verification_outputs`).

What. Walk donor trees under `data/source/HUBMAP/`, record every on-disk file (inventory), then separately check the **required teaching paths** for presence, measured size, and optional sha256. Write:

- `module1_file_inventory.tsv` / `module1_inventory_summary.tsv` (everything under donors)
- `module1_file_verification.tsv` (pass/fail on teaching cores)
- `module1_learner_download_manifest.tsv` and `module1_learner_download_summary.tsv` (course download budget)

Core required snRNA file per teaching block is `raw_expr.h5ad` plus block metadata `.tsv`. Optional files (`secondary_analysis.h5ad`, `expr.h5ad`, `scvelo_annotated.h5ad`) are inventoried when present but are not counted in the learner total.

Parameters: see Appendix B, section 2.

Why. A missing or truncated `raw_expr.h5ad` fails later as a shape or join error. Measuring sizes against the build record turns "I think I downloaded it" into an explicit pass/fail. Inventory can show gigabytes of optional blocks without changing the learner download total; the manifest is the checklist. If a file fails, replace it from the portal rather than adjusting the expected size to force a pass.

3. Provenance and access table

Script. `scripts/nb01_discovery/access_table.py` (`build_access_provenance_table`, `write_access_provenance_table`).

What. Emit `module1_dataset_access_provenance.tsv` with redistribution status **not redistributed; learner downloads from source portal**, plus roles for snRNA teaching blocks, Donor_2 multiome MOFA, and GTEx TPM used later.

Parameters: see Appendix B, section 3.

Why. The course package is code and derived tables/figures, not a data mirror. Access status belongs beside inventory so reviewers and learners see redistribution rules once.

4. Composition gate (airway-lineage epithelial labels)

Script. `scripts/nb01_discovery/composition_gate.py` (`donor_azimuth_crosstab`, `epithelial_airway_lineage_summary`, `write_composition_gate`). Label set `AIRWAY_LINEAGE_EPITHELIAL` (Club, Goblet, Deuterosomal, SMG, Tuft, multiciliated, transitional Club-AT2, and related). Backward-compatible alias `epithelial_gate` still calls the same epithelial summary; it is **not** a fifth coarse class.

What. Read `azimuth_label` (or fallback annotation columns) from each teaching block's `secondary_analysis.h5ad` in backed mode, crosstab labels by donor, and summarize epithelial airway-lineage membership before locking the cohort for Module 2. Outputs include:

- `module1_epithelial_airway_lineage_composition.tsv`
- `module1_epithelial_airway_lineage_by_donor.tsv`
- `module1_donor_azimuth_composition.tsv`
- `module1_azimuth_join_coverage.tsv`

This gate is a **pre-QC secondary analysis inventory**. It does not use Module 1 QC-filtered or subsampled objects. Module 2 composition and DE use post-QC, possibly subsampled integrated nuclei and map airway secretory / gland Azimuth labels into the coarse `'epithelial'` bucket only (four classes total: epithelial, endothelial, stromal, immune). Never treat a fifth coarse airway bucket as a Module 2 `cell_class`.

Parameters: see Appendix B, section 4.

Why. If teaching blocks lacked airway-lineage labels entirely, epithelial composition in Module 2 would be alveolar-only by construction. Classic SMG labels may still be rare; Club / transitional labels still contribute to epithelial. Donor_1 historically leaves more raw barcodes without labels than Donor_2/3; that is a join fact, not automatic QC failure.

5. QC filters

Script. `scripts/nb02_qc/qc.py` (`compute_qc_metrics`, `apply_filters`); load helpers in `scripts/nb02_qc/load.py`; per-block logs via `scripts/nb02_qc/export.py`.

What. For each teaching block, load `raw_expr.h5ad`, compute QC metrics (including mitochondrial and ribosomal percent), and apply filters in fixed order: gene prevalence, min genes per nucleus, min counts, max mitochondrial percent, max genes. Doublet calling is **not** applied on the core path (`module1.doublet_filtering: false`): inputs are already HuBMAP-processed nuclei. Each step is logged with nucleus and gene counts in `module1_qc_filter_log_<block_id>.tsv` and combined in `module1_qc_filter_log_all_donors.tsv`.

Parameters: see Appendix B, section 5.

Why. Defaults are transparent, reproducible, and matched to snRNA-seq rather than aggressive cell-line filters. Mitochondrial filtering at 5% is treated as a diagnostic ceiling: on these nuclear objects it often removes almost no nuclei, so depth and complexity dominate the filter path.

6. Preprocess, teaching subsample, neighbors, and UMAP

Script. `scripts/nb02_qc/preprocess.py` (`normalize_log`, `hvg_pca`, `maybe_subsample`, `neighbors_umap`); save via `scripts/nb02_qc/export.py` (`save_module1_outputs` / module1 h5ad writers).

What. After QC, run total-count normalization and `log1p`, select highly variable genes (Seurat flavor on log-normalized X), scale with clipping, and compute PCA. **Then** apply the teaching subsample (`module1.max_nuclei`) with `module1.random_seed`. **Only after** subsample compute neighbors and UMAP. Order matters: HVG/PCA see the full post-QC nucleus set; neighborhood graphs and UMAP see the capped teaching set. Preserve raw counts in `layers['counts']` for downstream DE. Write `data/processed/hubmap/module1_<block_id>_qc_preprocessed.h5ad` for Module 2.

Module 2 does **not** reuse this per-donor embedding for integration: it recomputes joint HVG/PCA (`module2.embedding.n_pcs=40`) on the concatenated object. Module 1 UMAP is a teaching QC view.

Parameters: see Appendix B, section 6.

Why. Module 2 needs a comparable preprocessed object per donor with a counts layer. The 5000-nucleus cap keeps classroom RAM predictable; Donor_1 often finishes QC below the cap and is not subsampled. Subsampling after HVG/PCA avoids re-estimating HVGs on a random subset while still reducing neighbor-graph cost.

Committed parameter stamp:

step	parameters	package_or_function
Input matrix	HuBMAP Salmon <code>raw_expr.h5ad</code> (counts-like)	anndata / scanpy
QC metrics	<code>scanpy.pp.calculate_qc_metrics</code> (mt, ribo)	scanpy
Gene filter	<code>min_cells=3</code>	<code>scanpy.pp.filter_genes</code>
Cell filter: complexity	<code>min_genes=200</code>	<code>scanpy.pp.filter_cells</code>
Cell filter: depth	<code>min_counts=500</code>	obs threshold
Cell filter: mito	<code>max_mito_percent=5.0</code>	obs threshold
Cell filter: max genes	<code>max_genes=8000</code>	obs threshold
Doublet filtering	Not applied in core path (HuBMAP processed nuclei; optional Scrublet deferred)	n/a
Normalization	<code>normalize_total target_sum=10000</code>	<code>scanpy.pp.normalize_total</code>
Log transform	<code>log1p</code>	<code>scanpy.pp.log1p</code>
Feature selection	Seurat HVG <code>n_top_genes=2000</code>	<code>scanpy.pp.highly_variable_genes</code>
Scaling	<code>max_value=10</code> (HVG subset)	<code>scanpy.pp.scale</code>
PCA	<code>n_comps=50</code> , <code>svd_solver=arpack</code>	<code>scanpy.tl.pca</code>
Neighbors graph	<code>n_neighbors=15</code> , <code>n_pcs=50</code>	<code>scanpy.pp.neighbors</code>

Showing 14 of 17 rows. Source: `module1_methods_parameters.tsv`

5. Results

Verification and download total

Verification covered 23 files: the 10 required teaching files, all present (`n_required_missing = 0`), and 13 optional files excluded from the budget. Every required row matches the manifest on `size_mb` and `sha256`, so nothing was truncated. Those optional rows measure 19,253.5 MB together, which is why the manifest, not the inventory, defines the download.

Module 1 verification covers the HuBMAP files it manages directly. The full course budget, including the GTEx, GEO and OSDR files that Module 4 requires, is itemized under "Option 1: minimum download, recommended" above, with retrieval commands in Appendix A.

The measured HuBMAP portion of the download is 394.60 MB of `raw_expr.h5ad` (68.57, 104.55, 150.92 and 70.56) plus 0.08 MB of block metadata TSVs, and the Donor_2 `multiome_mofa.hdf5` listed in the download

summary. With the GTEx, GEO and OSDR files that Module 4 requires, the full course download is reported in `module1_learner_download_summary.tsv`. Module 1 verification reports every required file as present and size-matched against the manifest.

QC filter path (all teaching donors)

donor_label	block_id	step	n_nuclei	n_genes	rationale
Donor_1	HBM347.VDHS.379	loaded	4899	98000	raw_expr.h5ad as QC starting matrix
Donor_1	HBM347.VDHS.379	min_cells_per_gene	4899	45859	Retain genes detected in ≥ 3 nuclei
Donor_1	HBM347.VDHS.379	min_genes	4346	45859	Remove low-complexity nuclei with < 200 genes
Donor_1	HBM347.VDHS.379	min_counts	3885	45859	Remove nuclei with < 500 total counts
Donor_1	HBM347.VDHS.379	max_mito_percent	3885	45859	Remove nuclei with mitochondrial percent > 5.0 (snRNA typically low mito; threshold is diagnostic)
Donor_1	HBM347.VDHS.379	max_genes	3884	45859	Remove extreme high-gene outliers > 8000
Donor_1	HBM347.VDHS.379	final_qc	3884	45859	Input for normalization / HVG / PCA / UMAP
Donor_1	HBM347.VDHS.379	teaching_subsample	3884	45859	no teaching subsample (n_nuclei=3884 $\leq$ module1.max_nuclei=5000)
Donor_4	HBM379.QCJT.435	loaded	7999	98000	raw_expr.h5ad as QC starting matrix
Donor_4	HBM379.QCJT.435	min_cells_per_gene	7999	47770	Retain genes detected in ≥ 3 nuclei
Donor_4	HBM379.QCJT.435	min_genes	7746	47770	Remove low-complexity nuclei with < 200 genes
Donor_4	HBM379.QCJT.435	min_counts	6282	47770	Remove nuclei with < 500 total counts
Donor_4	HBM379.QCJT.435	max_mito_percent	6278	47770	Remove nuclei with mitochondrial percent > 5.0 (snRNA typically low mito; threshold is diagnostic)
Donor_4	HBM379.QCJT.435	max_genes	6278	47770	Remove extreme high-gene outliers > 8000

Showing 14 of 32 rows. Source: `module1_qc_filter_log_all_donors.tsv`

Reading the table. 35,890 nuclei enter QC (4,899; 10,996; 11,996; 7,999 for Donors 1, 2, 3 and 4) and 28,948 survive to `final_qc` (3,884; 8,508; 10,278; 6,278), 77.4 to 85.7 percent per donor. Depth and complexity dominate: `min_genes` removes 829 cohort-wide and `min_counts` 6,105, while `max_mito_percent` at 5.0 removes 4 (all in Donor_4) and `max_genes` 4 more. The `teaching_subsample` row then caps Donor_2, Donor_3 and Donor_4 at `max_nuclei = 5000` with seed 0 and leaves Donor_1 at 3,884, already below the cap and logged as no subsample; 18,884 nuclei reach Module 2.

Azimuth join coverage

donor_label	n_raw	n_overlap	n_raw_without_label	fraction_raw_with_label
Donor_1	4899	4251	648.0	0.867728
Donor_2	10996	10989	7.0	0.999363
Donor_3	11996	11980	16.0	0.998666
Donor_4	7999	7657	342.0	0.957245

5 further columns in the source file. Source: module1_azimuth_join_coverage.tsv

34,877 of the 35,890 raw barcodes carry a secondary-analysis label, 97.2 percent, and all four joins ran. Donor_2 and Donor_3 are effectively complete (0.9994 and 0.9987; 7 and 16 barcodes missed), Donor_4 sits at 0.9572 (342 unlabeled) and Donor_1 at 0.8677 (648 of 4,899 unlabeled). Module 2 [cell_class](#) mapping inherits this join.

Airway-lineage composition gate

donor_label	block_id	airway_lineage_nuclei	labels_present	note
Donor_1	HBM347.VDHS.379	183	Basal resting; Club (non-nasal); Ionocyte; Multiciliated (non-nasal); Suprabasal; Transitional Club-AT2	composition summary only; not a DE group certificate
Donor_2	HBM473.NKMR.872	429	Basal resting; Ionocyte; Multiciliated (non-nasal); Suprabasal; Transitional Club-AT2	composition summary only; not a DE group certificate
Donor_3	HBM484.RGWP.797	622	Basal resting; Club (non-nasal); Ionocyte; Multiciliated (non-nasal); Transitional Club-AT2	composition summary only; not a DE group certificate
Donor_4	HBM379.QCJT.435	412	Basal resting; Club (non-nasal); Deuterosomal; Goblet (bronchial); Ionocyte; Multiciliated (non-nasal); Transitional Club-AT2	composition summary only; not a DE group certificate

Source: module1_epithelial_airway_lineage_composition.tsv

donor_label	block_id	Basal resting	Club (non-nasal)	Deuterosomal	Goblet (bronchial)	Ionocyte	Multiciliated (non-nasal)
Donor_1	HBM347.VDHS.379	20	15	0	0	3	55
Donor_2	HBM473.NKMR.872	3	0	0	0	167	171
Donor_3	HBM484.RGWP.797	42	4	0	0	74	226
Donor_4	HBM379.QCJT.435	2	25	2	2	33	295

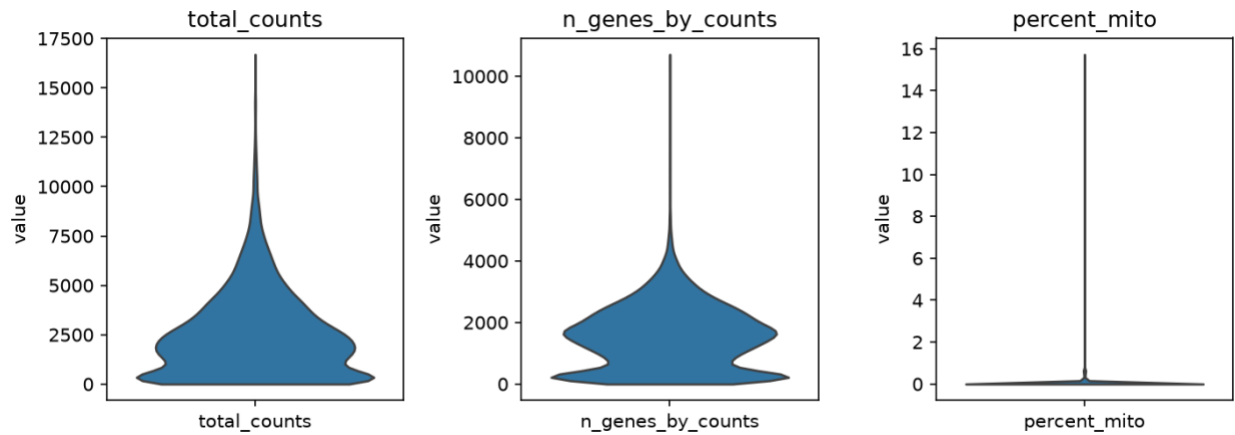
2 further columns in the source file. Source: module1_epithelial_airway_lineage_by_donor.tsv

Reading the gate. The gate passes: all four donors contribute airway-lineage nuclei, 1,646 across the cohort (183, 429, 622 and 412 for Donors 1, 2, 3 and 4) over eight labels. Multiciliated (non-nasal) leads at 747, then Transitional Club-AT2 at 505 and Ionocyte at 277, each in every donor. The tail is thin: Basal resting 67, Club (non-nasal) 44, Suprabasal 2, Deuterosomal and Goblet (bronchial) 2 each and only in Donor_4, and no

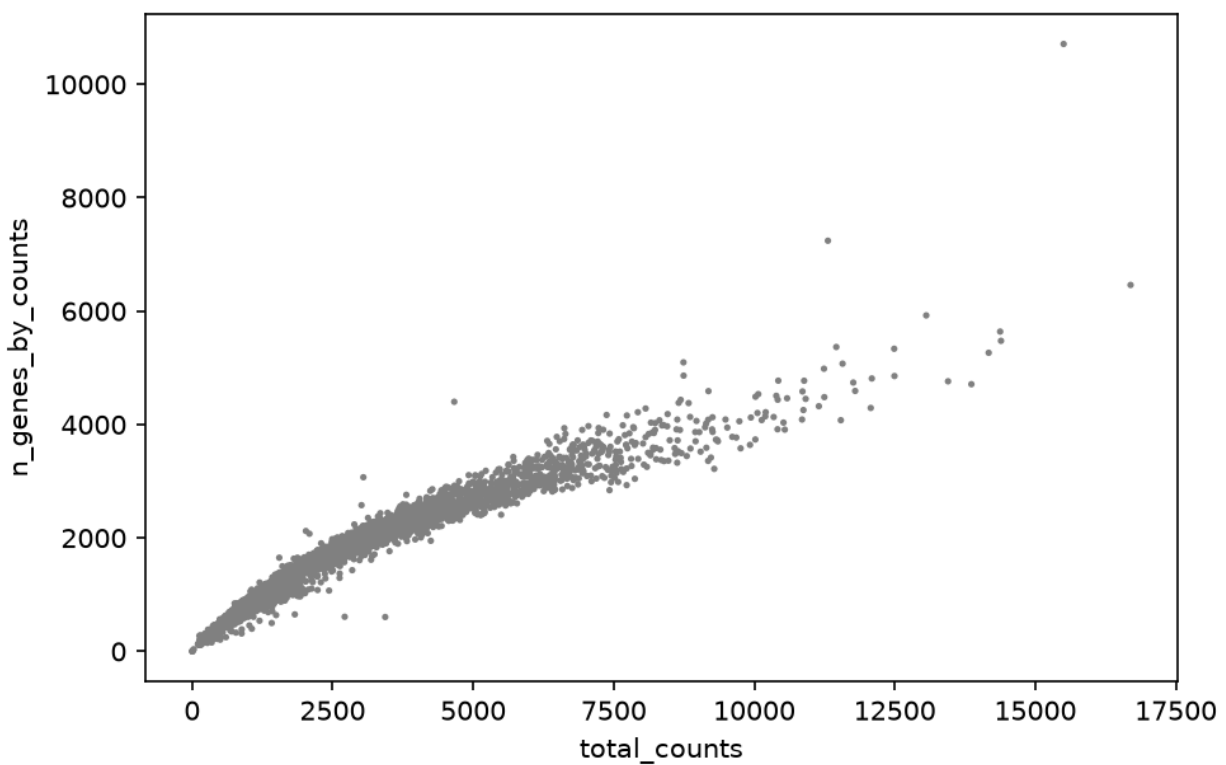
submucosal gland label. Module 2 therefore gets airway-aware **epithelial** membership, thin outside the top three labels, with no fifth coarse airway class.

QC figures (Donor_1 teaching scaffold)

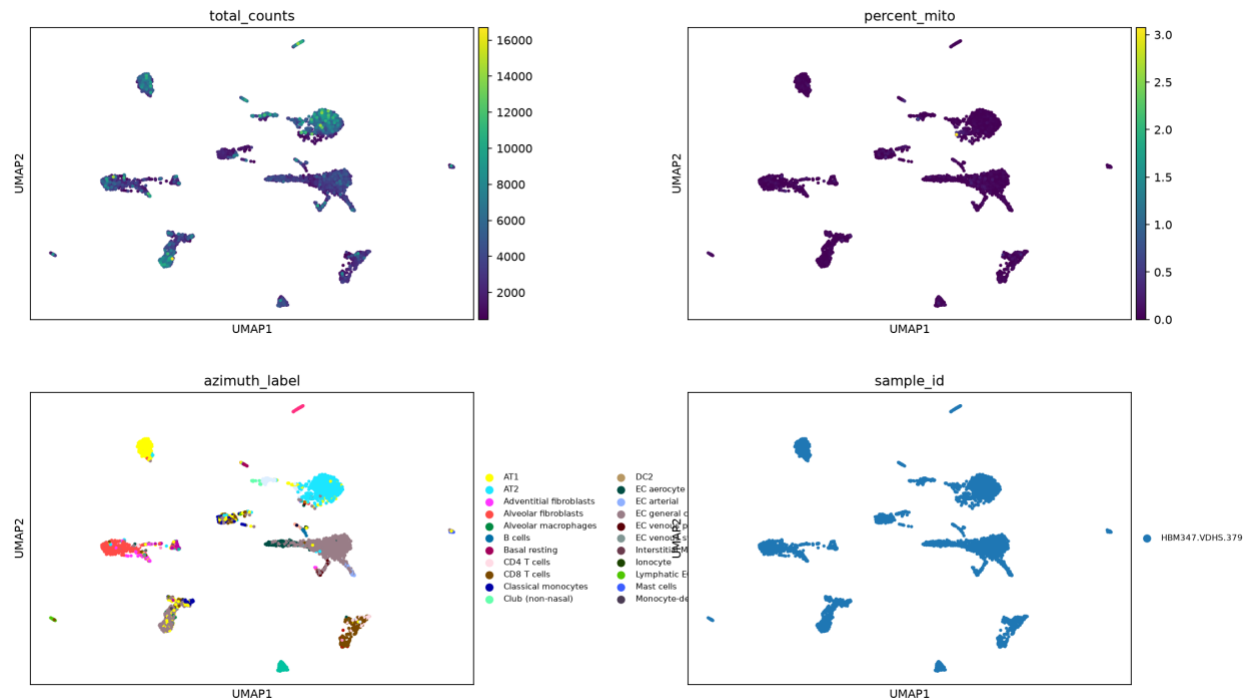
Donor_1 is the notebook's single-block dry-run; the same plots exist for every teaching block under [outputs/figures/](#).



Violin plots of QC metrics for Donor_1 teaching block HBM347.VDHS.379



Total counts versus genes detected for Donor_1 after QC metrics calculation



UMAP of QC-preprocessed nuclei for Donor_1 teaching block HBM347.VDHS.379

6. Interpretation

Module 1 establishes a verified local teaching set and a documented QC trail. The scientific claim at this stage is modest and procedural: the four HuBMAP lung snRNA blocks required for Module 2 are present, size-checked, composition-gated, and preprocessed under explicit thresholds.

What the results support.

- A learner can reconstruct which files are required for the whole course and where they belong on disk from this companion alone.
- QC logs show which filters remove nuclei; preprocess objects are ready for Harmony integration.
- Primary DOI vs processed download ID is an operational fact of HuBMAP access, not a paperwork detail.
- Airway-lineage labels are present across donors at teaching-relevant counts, with donor-to-donor imbalance that Module 2 must not over-interpret.

Interpretation of the QC pattern. Depth and complexity filters dominate removals; mitochondrial filtering at 5% is nearly inactive on these snRNA objects. That pattern matches nuclear RNA expectations and is why the mito threshold is described as diagnostic rather than as the main cleaning lever. The teaching subsample then equalizes RAM: three donors are capped at 5000 nuclei while Donor_1 remains at its post-QC count when already below the cap.

Interpretation of label join. Incomplete Azimuth join on Donor_1 (hundreds of raw barcodes without labels) means composition summaries that require labels are conditioned on the overlapping barcode set. Reporting `module1_azimuth_join_coverage.tsv` keeps that conditioning visible when Module 2 maps `cell_class`.

7. Extending this analysis

Each item below states what a stronger version of this work would require to support a claim Module 1 leaves open, and what the current evidence does support.

- **Cohort selection by live query.** Showing that a live Search API query selects this cohort would mean running the query at teaching time and recording the returned IDs beside the curated list; `module1.discovery.live_query` (default `false`) turns that check on. This cohort was curated for teaching value, so the two lists have not yet been compared.
- **Donor integration and biological contrasts.** Testing differences across age or disease needs a joint embedding and a DE model, which is exactly what Module 2 builds on these QC objects and Module 4 extends across ecosystems. Module 1 stops at per-donor objects on purpose; the next step is Module 2 concatenation.
- **A powered donor-age contrast** would need many more donors per arm. This four-donor split (younger Donor_2 and Donor_1 versus older Donor_4 and Donor_3) is a design illustration of how an age variable enters the design, and the next step is a larger cohort pulled through the same HuBMAP access path.
- **Label quality.** Treating Azimuth labels as ground truth would require an independent annotation, marker-based or expert, to compare against. Until that exists, `module1_azimuth_join_coverage.tsv` travels with the labels so any label-conditioned count can be discounted by the barcodes that never received one (648 of 4,899 in Donor_1).
- **Blocks outside the manifest.** Bringing an extra HuBMAP block from `data/source/` into the analysis means adding it to `module2.inputs`, then re-running verification, the composition gate, and QC before Module 2; that is the same path used to widen the cohort. Until then those blocks are inventoried only, which is what keeps the download budget stable.

Forward link. Module 2 concatenates these QC objects, measures donor mixing with Harmony, and runs within-HuBMAP DE and pathway enrichment. GTEx, GEO, and OSDR appear in Module 4; Donor_2 multiome (`multiome_mofa.hdf5`) is interpreted in Module 3.

8. Open questions

Use the Module 1 tables and figures already written under `outputs/`.

1. Which QC threshold removed the most nuclei, and would a neighboring cut change the donor composition story?
2. What does Azimuth join coverage tell you about unlabeled barcodes on Donor_1 versus Donor_4?
3. What claim about airway lineages is **not** supported by this cohort after QC?
4. If a colleague treated HubMAP processed IDs as interchangeable with primary DOIs, what would go wrong?

Appendix A: obtaining the files from a terminal

The screencast walks the same acquisition through each portal's web interface. This appendix is the equivalent from a terminal, for a learner who prefers a scripted, repeatable pull. Every command was executed against the live endpoints and every size confirmed against the download manifest.

Run these from the package root. Each command writes into the folder `config/paths.yaml` expects, so no file needs moving afterwards.

A.1 Finding the APIs: the SmartAPI registry

The registry is the entry point. This returns the registered HuBMAP entries, including the Search and Entity APIs used below:

```
curl -s "https://smart-api.info/api/query?q=hubmap&fields=info.title,info.version,servers"
```

A.2 Resolving a HuBMAP identifier to an asset UUID

Asset URLs are addressed by UUID, not by the HuBMAP identifier you see in a portal. The Entity API performs the translation. For the Donor_1 processed release:

```
curl -s https://entity.api.hubmapconsortium.org/entities/HBM675.RVGB.258 \
| python -c "import json,sys; print(json.load(sys.stdin)['uuid'])"
```

The same call on a primary block returns `data_access_level: protected`, which is the access tier distinction above expressed as a field you can test in code.

A.3 HuBMAP matrices and the multiome object

```
mkdir -p data/source/HUBMAP/Donor_1/snRNAseq/HBM347.VDHS.379 \
data/source/HUBMAP/Donor_2/snRNAseq/HBM473.NKMR.872 \
data/source/HUBMAP/Donor_3/snRNAseq/HBM484.RGWP.797 \
data/source/HUBMAP/Donor_4/snRNAseq/HBM379.QCJT.435 \
data/source/HUBMAP/Donor_2/SNAREseq/HBM828.GPVG.252

BASE=https://assets.hubmapconsortium.org
curl -L -o data/source/HUBMAP/Donor_1/snRNAseq/HBM347.VDHS.379/raw_expr.h5ad \
$BASE/171bd6114652ce2a1ff0e4757b5eacf9/raw_expr.h5ad
curl -L -o data/source/HUBMAP/Donor_2/snRNAseq/HBM473.NKMR.872/raw_expr.h5ad \
$BASE/4c031c4aba3b656a07a5ab14a305cbba/raw_expr.h5ad
curl -L -o data/source/HUBMAP/Donor_3/snRNAseq/HBM484.RGWP.797/raw_expr.h5ad \
$BASE/ab3942cee59a994d73f6f89f8f5eb84a/raw_expr.h5ad
curl -L -o data/source/HUBMAP/Donor_4/snRNAseq/HBM379.QCJT.435/raw_expr.h5ad \
$BASE/f5ce0bfcef8d17a18e0a3911fb2b2726/raw_expr.h5ad
curl -L -o data/source/HUBMAP/Donor_2/SNAREseq/HBM828.GPVG.252/multiome_mofa.hdf5 \
$BASE/772d0ce4657bb306359c0230c57ee1e9/multiome_mofa.hdf5
```

A.4 The four block metadata TSVs: portal interface only

These four files have no scriptable URL. The pipeline reads them in HuBMAP's Key/Value export format, with columns **HuBMAP ID**, **Entity**, **Key** and **Value**, and `scripts/common/io.py` raises if any is absent. The programmatic metadata endpoints return a different, wide layout that will not parse.

Obtain each from the portal instead, at <https://portal.hubmapconsortium.org/browse/dataset/<primary block ID>>, using the metadata download control on the dataset page, and save it beside that donor's `raw_expr.h5ad` under the primary block ID with a `.tsv` extension:

Donor	Portal page	Save as
-------	-------------	---------

Donor_1	HBM347.VDHS.379	Donor_1/snRNAseq/HBM347.VDHS.379/HBM347.VDHS.379.tsv
Donor_2	HBM473.NKMR.872	Donor_2/snRNAseq/HBM473.NKMR.872/HBM473.NKMR.872.tsv
Donor_3	HBM484.RGWP.797	Donor_3/snRNAseq/HBM484.RGWP.797/HBM484.RGWP.797.tsv
Donor_4	HBM379.QCJT.435	Donor_4/snRNAseq/HBM379.QCJT.435/HBM379.QCJT.435.tsv

That a public repository serves most of its files at stable URLs and one export only through a browser is worth noticing rather than working around. Automating an acquisition means finding these boundaries, and they are not documented anywhere except by trying.

A.5 GTEX

```
mkdir -p data/source/GTEX
GTEX=https://storage.googleapis.com/adult-gtex
curl -L -o data/source/GTEX/gene_tpm_v10_lung.gct.gz \
  $GTEX/bulk-gex/v10/rna-seq/tpms-by-tissue/gene_tpm_v10_lung.gct.gz
curl -L -o data/source/GTEX/gene_reads_v10_lung.gct.gz \
  $GTEX/bulk-gex/v10/rna-seq/counts-by-tissue/gene_reads_v10_lung.gct.gz
curl -L -o data/source/GTEX/GTEX_Analysis_v10_Annotations_SampleAttributesDS.txt \
  $GTEX/annotations/v10/metadata-files/GTEX_Analysis_v10_Annotations_SampleAttributesDS.txt
curl -L -o data/source/GTEX/GTEX_Analysis_v10_Annotations_SubjectPhenotypesDS.txt \
  $GTEX/annotations/v10/metadata-files/GTEX_Analysis_v10_Annotations_SubjectPhenotypesDS.txt
```

A.6 GEO

```
mkdir -p data/source/GEO/GSE150910
GEO=https://ftp.ncbi.nlm.nih.gov/geo/series/GSE150nnn/GSE150910
curl -L -o data/source/GEO/GSE150910/GSE150910_gene-level_count_file.csv.gz \
  $GEO/suppl/GSE150910_gene-level_count_file.csv.gz
curl -L -o data/source/GEO/GSE150910/GSE150910_series_matrix.txt.gz \
  $GEO/matrix/GSE150910_series_matrix.txt.gz
```

A.7 NASA OSDR

OSDR serves files through a study-scoped download endpoint. The ISA metadata arrives as one archive whose members are flat, so it must be expanded into the directory name the config resolves:

```
mkdir -p data/source/OSDR/OSD-248
OSDR="https://osdr.nasa.gov/geode-py/ws/studies/OSD-248/download?source=datamanager&file="
cd data/source/OSDR/OSD-248
curl -L -o GLDS-248_rna_seq_RSEM_Unnormalized_Counts_rRNArm_GLbulkRNAseq.csv \
  "${OSDR}GLDS-248_rna_seq_RSEM_Unnormalized_Counts_rRNArm_GLbulkRNAseq.csv"
curl -L -o GLDS-248_rna_seq_differential_expression_rRNArm_GLbulkRNAseq.csv \
  "${OSDR}GLDS-248_rna_seq_differential_expression_rRNArm_GLbulkRNAseq.csv"
curl -L -o GLDS-248_rna_seq_contrasts_GLbulkRNAseq.csv \
  "${OSDR}GLDS-248_rna_seq_contrasts_GLbulkRNAseq.csv"
curl -L -o isa.zip "${OSDR}OSD-248_metadata_OSD-248-ISA.zip"
unzip -o isa.zip -d OSD-248_metadata_GLDS-248-ISA && rm isa.zip
cd -
```

The archive is named `OSD-248_metadata_OSD-248-ISA.zip` while `config/paths.yaml` resolves `OSD-248_metadata_GLDS-248-ISA/`, which is why `unzip -d` names the destination explicitly. To see what a study holds before choosing files, <https://osdr.nasa.gov/osdr/data/osd/files/248> returns the full listing as JSON.

A.8 Confirming what you retrieved

```
python scripts/verify_access.py
```

One caution, and it is the reason this course asks you to date-stamp a measurement. Portal files are republished. The OSD-248 ISA assay table measured 0.16 MB on the authoring run and measures 0.22 MB today, so a total

recomputed now will not match the committed manifest to the last decimal. That is the repository behaving normally, not an error in your download. A size that is wildly different, or a file that is absent, is a real failure; a small drift in a metadata table is provenance, and the right response is to record what you measured and when.

Appendix B: parameters

Config keys and defaults for each Methods step. Section numbers match the Methods subsections; values are the live entries in `config/paths.yaml` under `module1` (and related keys).

1. Identify and place teaching files

Config key	Default	Definition	If modified	Impact on results
<code>module1.donor_label</code>	Donor_1	Notebook dry-run donor for single-block QC figures	Switch donor	Changes which block is used for Module 1 scaffold plots only
<code>module1.block_id / primary_id</code>	HBM347.VDHS.379	Primary teaching block ID and folder name	Pin another primary	New matrix and barcodes; full Module 1+2 rebuild
<code>module1.processed_id</code>	HBM675.RVGB.258	Public processed release actually downloaded	Pin different descendant	Different bytes and possibly barcode set; re-verify sizes
<code>module1.primary_doi</code>	10.35079/HBM347.VDHS.379	Citation DOI for the primary	Editorial only	Does not change analysis matrices
<code>module2.inputs[*]</code>	four donor blocks	Full integration cohort Module 1 verifies	Add/drop a donor	Changes inventory, QC objects, and all Module 2 counts

2. Inventory and measured-size verification

Config key	Default	Definition	If modified	Impact on results
<code>paths.hubmap_source</code>	<code>data/source/HUBMAP</code>	Root for donor trees	Relocate tree	Inventory and verification paths break until files move
<code>module2.inputs</code>	four blocks	Teaching set used by <code>teaching_inputs()</code>	Change cohort	Manifest rows and verification targets change
<code>module1.discovery.live_query</code>	false	Live SmartAPI / Search API (deferred)	Set true	Optional registry tables; does not replace local verify
<code>module1.discovery.cache_registry</code>	<code>outputs/tables/module1_smartapi_registry.tsv</code>	Offline cache path for registry	Path change	Affects discovery cache only

3. Provenance and access table

Config key	Default	Definition	If modified	Impact on results
<code>module1.primary_doi / per-donor primary_doi</code>	see cohort block	DOI cited for each primary	Editorial	Provenance text only
<code>module3.dataset_id</code>	HBM828.GPVG.252	Multiome product listed for access	Change product	Provenance row for Module 3 input
<code>module4.data.gtex_tpm</code>	<code>data/source/GTEX/gene_tpm_v10_lung.gct.gz</code>	GTEx TPM path in access map	Relocate	Provenance path string; Module 4 load must match

4. Composition gate (airway-lineage epithelial labels)

Config key	Default	Definition	If modified	Impact on results
module1.join_azimuth_labels	true	Whether QC export joins labels onto barcodes	Set false	Join coverage / labeled composition thin out
module2.labels.fetch_remote	false	Live HTTP range-read of <code>azimuth_label</code> from public secondary h5ad	Set true	May change join coverage vs local secondary
module2.labels.cache	outputs/tables/module2_azimuth_labels.tsv	Cached label TSV for Module 2	Refresh cache	Label joins for Module 2 change
module2.label_column	azimuth_label	Obs column used for coarse mapping later	Rename	Must match joined column or mapping fails

5. QC filters

Config key	Default	Definition	If modified	Impact on results
module1.qc.min_cells_per_gene	3	Drop genes detected in fewer nuclei	Raise	Fewer genes enter preprocess / Module 2 intersection
module1.qc.min_genes	200	Drop low-complexity nuclei	Raise	Removes more nuclei; rare lineages shrink first
module1.qc.min_counts	500	Drop shallow nuclei by total counts	Raise	Same direction as <code>min_genes</code> ; depth-driven loss
module1.qc.max_mito_percent	5.0	Mitochondrial percent ceiling	Lower further	Rarely removes more on snRNA; raise admits damaged nuclei
module1.qc.max_genes	8000	Drop extreme high-complexity nuclei	Lower	Can remove legitimate high-complexity epithelial nuclei
module1.doublet_filtering	false	Optional doublet call (off)	Set true	Extra removals; not the teaching default

6. Preprocess, teaching subsample, neighbors, and UMAP

Config key	Default	Definition	If modified	Impact on results
module1.preprocess.target_sum	10000	<code>normalize_total</code> target	Change scale	Rescales log expression; markers shift mildly
module1.preprocess.n_top_genes	2000	HVG count (Seurat flavor)	Raise/lower	Changes PCA basis for Module 1 UMAP only
module1.preprocess.n_pcs	50	PCA components stored on object	Change	Module 1 neighbors/UMAP geometry; Module 2 still recomputes

module1.preprocess.n_neighbors	15	kNN for Module 1 UMAP	Change	Local Module 1 UMAP structure
module1.preprocess.umap_min_dist	0.3	UMAP min_dist	Lower	Tighter Module 1 clusters in the plot
module1.preprocess.umap_spread	1.0	UMAP spread	Raise	More dispersed Module 1 UMAP
module1.preprocess.scale_max_value	10	Clip after scaling before PCA	Lower	Compresses outlier gene influence on Module 1 PCA
module1.max_nuclei	5000	Teaching cap after HVG/PCA	Raise	More nuclei into Module 2; higher RAM
module1.random_seed	0	RNG for subsample and neighbors/UMAP	Change seed	Different subsample draws and UMAP stochasticity

References

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6. Wolf FA, Angerer P, Theis FJ. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol*. 2018;19(1):15. PMID: [29409532](#).
7. McInnes L, Healy J, Melville J. UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. *arXiv*. 2018. arXiv: [1802.03426](#).
8. HuBMAP primary dataset DOIs for this teaching set: [10.35079/HBM347.VDHS.379](#), [10.35079/HBM473.NKMR.872](#), [10.35079/HBM484.RGWP.797](#), [10.35079/HBM379.QCJT.435](#).